Mode-matched record horizons in reversible dynamics: relaxation, redundancy, and protected exceptions

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In the standard account of the arrow of time the memory arrow is slaved to the thermodynamic gradient: records form on the entropy-increasing side of a low-entropy boundary condition. Here we quantify record-keeping itself. In an exactly bit-reversible lattice gas we place two clocks on the same runs: the entropy-saturation time $t_{\rm S}$ (coarse-grained entropy reaches $0.9\,S_{\rm max}$) and an *operational record horizon* t^*, the time at which an explicit held-out linear readout of the present coarse state retains less than half a bit about a planted macroscopic fact. For boundary-scale records we find $t^* = \kappa\, t_{\rm S}$ with κ of order one and flat across scrambling rates ($\kappa = 1.08$ over a $7\times$ range) and system sizes ($\bar{\kappa} = 0.99$, $L = 48$–128); the same proportionality reproduces in a continuous hard-disk gas ($\kappa = 1.00$) and in exactly reversible Clifford circuits of up to 128 qubits ($\kappa = 0.79$), and survives a Haar-random (non-Clifford) control. A one-slow-mode analysis accounts for the law: entropy deficit and record signal relax on a shared timescale, so both crossings take the form $\tau \ln(A/\theta)$ and κ is a ratio of logarithms; the reconstruction has no additional free parameters (median error 5%/1%) and predicts held-out runs at comparable accuracy. The analysis' sharper consequence is confirmed by making the record's wavelength an independent knob: the entropy clock is unchanged ($t_{\rm S}$ varies by 2%) while the horizon spans $9\times$, each mode's lifetime given by its own $\tau_M \ln(A_M/\theta)$ to 1–5%. *A record lives as long as the mode that carries it*; among relaxing records the longest horizons are of order the thermalization time, attained at the boundary scale, while a non-relaxing mode escapes the clock entirely. The law fails exactly where it must—an engineered conservation law decouples t^* from $t_{\rm S}$, and one of 25 lattice realizations does the same *spontaneously*, storing its record in a diagnosed frozen cluster at full fidelity for $48\,t_{\rm S}$; the incidence of this frozen sector is itself measured, closed at weak disorder (0/448 worlds) and opening sharply (5.5%, then 43%) at the strongest scrambling rates—and records are stored redundantly, with effective redundancy growing as N^{α}, $\alpha_\infty = 0.92 \pm 0.04$ (conservative) classically and $\alpha = 1$ for quantum broadcast states. These results quantify the sense in which the recorded past is the direction of the entropy gradient—while leaving the low-entropy boundary itself unexplained.

I. INTRODUCTION

Microscopic dynamics—classical Hamiltonian flow, unitary quantum evolution, or the exactly bit-reversible cellular automata used below—has no arrow of time. The observed asymmetry is standardly traced to a low-entropy boundary condition (the Past Hypothesis) together with coarse-graining [1–5]. Within this picture the *memory* (or "psychological") arrow is derivative: a physical record is a present macrostate correlated with a macrostate at another time, and such correlations are abundant only on the entropy-increasing side of the boundary [6–11]. Reichenbach's fork asymmetry, Wolpert's analysis of memory systems, and Mlodinow and Brun's argument that a robust record-forming system must sit on the entropy-increasing side all support the same conclusion: *memory points down the entropy gradient.*

The quantitative structure of record-keeping is less settled. Rovelli has argued that individual memories degrade as the memory system thermalizes [9]; Riedel, Zurek, and Zwolak showed that relaxation toward equilibrium destroys the *redundancy* of environmental records in decoherence models [12]; and the quantum Darwinism program [13–16] identified redundant imprinting as what makes a fact objective while it lasts. Observational entropy [17–20] supplies the bookkeeping for coarse observers of closed systems, and its increase has recently been shown to be generic in isolated systems [21]. What these threads have not provided is a controlled, same-run comparison of a record's operational lifetime against the thermodynamic clock of the same dynamics; a test of *what* sets that lifetime when the record's spatial structure is varied; and the boundary of the resulting law—the regime in which it demonstrably fails.

This paper reports such measurements. Our testbed is deliberately minimal: exactly reversible model dynamics (a Margolus block cellular automaton [22, 23], an event-driven hard-disk gas, and Clifford circuits [24, 25]), a coarse-grained Boltzmann entropy, and a one-bit macroscopic fact planted at the low-entropy boundary whose readability we track with an explicit decoder. Our contributions:

1. **Bookkeeping** (Sec. II): over an ensemble of microstates evolved by a bijective map, the fine-grained (Gibbs/Shannon) entropy is conserved to machine precision while the observational entropy rises; the rise is *identically* the growth of information hidden from the coarse observer, $\Delta S_{\rm obs} = \Delta I$.

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In parallel runs, a planted record decays to zero as the same coarse description saturates.

2. **A horizon law for boundary-scale records** (Sec. III): the record horizon tracks the entropy-saturation time, $t^* = \kappa\, t_S$ with κ order one, flat across scrambling rates ($7\times$ in t_S) and system sizes.

3. **A one-slow-mode account** (Sec. IV): entropy deficit and record signal decay exponentially on a shared timescale, making both crossings $\tau \ln(A/\theta)$ and κ a ratio of logarithms. The account reconstructs the measured crossings with no additional free parameters (median error 5%/1%) and predicts held-out runs out-of-sample at comparable accuracy.

4. **The mode-resolved law** (Sec. V): making the record's wavelength an independent knob shows the general statement is *mode-matched*: the entropy clock does not care which record was planted (t_S flat to 2%) while the horizon spans $9\times$, each mode's lifetime reproduced by its own $\tau_M \ln(A_M/\theta)$ to 1–5%. "Record lifetime = thermalization time" is the *supremum* of this law over relaxing records, attained by those written in the slowest mode.

5. **Cross-substrate corroboration** (Sec. VI): the boundary-scale proportionality reproduces in a continuous hard-disk gas and in reversible Clifford circuits (up to $N = 128$ qubits), and survives a Haar-random statevector control.

6. **A falsification boundary, with a measured incidence** (Secs. VII, III): the law is contingent on ergodicity. A conservation law protecting the record makes t^* diverge while t_S stays finite—the extreme case of a non-relaxing mode—and the spontaneous counterpart, the kinetically frozen record, has a sharp disorder-driven onset: closed at weak disorder (0/448 worlds), 5.5% at scatter 0.42, 43% at 0.50.

7. **Redundancy scaling** (Sec. VIII): near the boundary the record is recoverable from any $\sim 1/8$ of the environment (effective redundancy $R_\delta \approx 8$–10); R_δ grows with the number of environment fragments as N^α with a conservative $\alpha_\infty = 0.92 \pm 0.04$, while a quantum broadcast realizes ideal Darwinism, $\alpha = 1$ exactly—consistent with the classical shortfall being a signal-to-noise and finite-size effect rather than a limit of the principle.

Throughout we are explicit about scope and about what is *not* shown (Sec. IX): the low-entropy boundary is an input, as it must be for any time-symmetric dynamics [26]; t^* is an operational, decoder-relative quantity (a lower bound on the accessible-information horizon), and the numerical value of κ is convention-dependent—the physics is the shared relaxation and its mode structure. All code, the scripts generating every figure, and per-run data for the core experiments are public [27].

II. SETUP: SUBSTRATES, RECORDS, AND BOOKKEEPING

A. Reversible substrates and coarse entropy

Our primary substrate is a Margolus block cellular automaton [22, 23]: a two-dimensional lattice gas on an $L \times L$ grid of binary site occupations, updated in 2×2 blocks by bijective lookup tables with alternating block offsets. A quenched random fraction ("scatter") of blocks applies a rotation rule, breaking spurious conservation laws so the gas thermalizes cleanly; the map remains exactly bijective, and the inverse map recovers the initial bitstring exactly (verified bit-for-bit [27]). Appendix A gives details. Two companion substrates are used for universality tests: an event-driven hard-disk gas (elastic disks, specular walls; exact between-collision flight) and Clifford brickwork circuits simulated in the stabilizer formalism (Appendix C).

The coarse-grained (Boltzmann) entropy is defined by partitioning the grid into $b \times b$ cells and counting microstates compatible with the cell occupancies $\{n_i\}$:

$$S \;=\; \sum_i \ln \binom{b^2}{n_i}, \tag{1}$$

with $S_{\max}$ its microcanonical maximum at the given particle number. Initial states are low-entropy "blobs": all particles confined to a region of width w at density ρ, with microstates drawn microcanonically within the blob.

B. Records and the two clocks

A *record* is operationalized as follows. A one-bit macroscopic fact F (in the baseline construction, a marker of identical particle number placed on the left or the right side of the blob, so the two classes differ in no conserved quantity) is planted at $t = 0$. For each class we evolve an ensemble of K microstates through the same quenched disorder. The readout at time t is a nearest-centroid *linear* decoder $\hat{F}$ trained on the coarse occupancy vectors of half of the worlds and evaluated on the held-out half; the accessible record is the mutual information $I(F; \hat{F})$ estimated from the held-out confusion matrix (Appendix B). This is a *present-macrostate* readout: the decoder sees only the coarse state at time t, never the history.

Two caveats define the scope of t^*. First, it is *operational*: a particular readout's horizon, hence a lower bound on the horizon of the full accessible information $I(F; \text{coarse state})$. Section V quantifies the readout dependence: at the slowest mode three different readouts (trained linear, untrained fixed-template, debiased Gaussian plug-in) agree to within the logarithmic threshold offsets the mechanism predicts, while non-adaptive readouts of *drifting* fine-scale patterns lose the record earlier for physical reasons. Second, the thresholds below are conventions; Sec. IV shows crossings depend on them only logarithmically, and the time-rescaled collapses make the convention-independent content explicit.

The two clocks, defined on the same runs:

$$t_S: \quad \text{first time } S(t) \geq 0.9\, S_{\max}, \tag{2}$$

$$t^*: \quad \text{first time } I(F;\hat{F}) < \tfrac{1}{2} \text{ bit}. \tag{3}$$

C. Bookkeeping: the ledger identity

Before measuring lifetimes we fix the bookkeeping. Consider an ensemble of K equiprobable microstates evolved by the (bijective) dynamics. The Shannon entropy H of the microstate distribution is exactly conserved, $H = \ln K$; in our runs ($K = 64$, $L = 128$, $T = 2400$ steps) the deviation is zero to machine precision, $\max_t |H - \ln K| = 0$. The observational entropy [17, 18] of the coarse observer decomposes as

$$S_{\mathrm{obs}}(t) \;=\; H \;+\; I(t), \qquad I(t) \geq 0, \tag{4}$$

where $I(t)$ is the information about the microstate hidden from the coarse observer (Appendix B). Equation (4) is an identity of the coarse-graining, not an empirical finding; its content here is that H is exactly constant under the bijective dynamics, so *all* growth of S_{obs} is growth of the hidden-information term (Fig. 1, left). In parallel runs of the same substrate class ($K = 64$, $L = 64$), a planted record decays from 1.000 bit to 0.006 bit as the coarse entropy saturates (Fig. 1, right). We present these as two faces of one coarse-graining—the observer's global ignorance grows while its grip on any planted fact decays—without claiming a computed decomposition of $I(t)$ into record-specific parts; the quantitative link between the two clocks is the subject of the rest of the paper.

III. THE HORIZON LAW FOR BOUNDARY-SCALE RECORDS

Figure 2 shows the baseline measurement. We sweep the scrambling rate (scatterer fraction 0.15–0.42) with per-condition run lengths chosen so that no condition is clamped by the simulation window ($T = 400$–2400 steps, sampling stride 2–6), five quenched-disorder seeds per condition, $K = 40$ worlds per fact class, $L = 64$, $b = 4$. Of 25 runs, 24 resolve both clocks; one run at the slowest condition remains censored.

The resolved runs give a proportionality through the origin,

$$t^* \;=\; 1.08\, t_S, \qquad R^2 = 0.82, \tag{5}$$

over a 7× range of t_S (26–192 steps). More important than the fit is its *flatness*: the per-condition ratios are 1.37, 0.97, 1.09, 1.02, 1.15 (standard deviations 0.21–0.28), so the law holds run by run rather than through one leverage point. (The largest ratio occurs at the fastest condition, where both times are shortest and the discrete sampling is coarsest relative to them.) A condition-stratified bootstrap gives a 95% confidence interval $\kappa \in [0.91, 1.24]$; an unconstrained affine fit $t^* = a + b\, t_S$ yields an intercept consistent with zero ($a = 2.5$, 95% CI $[-16.2, 20.4]$) and is not preferred over the one-parameter proportionality (AIC 154.4 vs 156.3). The one censored run is qualitatively different, and we diagnose rather than merely disclose it: its record never degrades at all—$I(F;\hat{F})$ dips to 0.855 bit and returns to 1.0, remaining perfectly readable when the run is extended to $t = 12\,000 \approx 48\, t_S$, while its four sibling seeds cross at $t^* \approx 192$ (Fig. 3). The diagnosis: the planted marker is a solid patch, and a fully occupied (or fully empty) 2×2 block is a fixed point of every block rule *within a single partition*—under the alternating partitions this makes the *interiors* of solid and void clusters inert while their boundaries can still erode, and in this one quenched world the erosion stalls. At $t = 2400$, 87% of the class-discriminating signal sits inside the marker boxes and 81% of it on sites that never move over the measurement window; those frozen sites have occupancy 0.93 in each class's own marker box (a solid remnant) and 0.07–0.15 in the opposite box (a void remnant). The realization is non-ergodic where the record lives *on every timescale we can observe*: the record's mode shows no relaxation out to $t^* \geq 12\,000 \approx 48\, t_S$—a kinetically frozen record, consistent with (though, absent an invariance proof, not established to be) the protected $\tau_{\mathrm{mode}} \to \infty$ limit of Eq. (10) and of Sec. VII, arising here spontaneously from quenched disorder rather than by engineering. We therefore state the horizon law as holding in 24/25 realizations, with the exception classified as a frozen-record sector rather than folded into the fit (as a point value at its lower bound $t^* \geq T$ it would shift the through-origin fit to 3.0 through a single high-leverage point); all its lifetime statements are lower bounds.

Is the frozen realization a fluke? We measured its *incidence* (Fig. 4): 798 additional independent quenched worlds across six scrambling rates, each evolved to $T \approx 8\, t_S$ and classified by the same operational readout (frozen: late-time MI ≥ 0.8 bit; partial/slow: $[0.3, 0.8)$ bit, reported separately and never folded in). The frozen sector has a sharp, disorder-driven onset. It is closed at scatter 0.15–0.35 (0 of 448 worlds; 95% upper bounds 2.5–3.8%), opens at 0.42 (11/200 = 5.5%, CI $[3.1, 9.6]$%—consistent with the 1/25 of the horizon sweep), and reaches 43.3% (65/150, CI $[35.7, 51.3]$%) at scatter 0.50; the partial/slow population tracks the onset ($0 \to 3 \to 29 \to 57$ worlds), and at the strongest disorder 81% of worlds retain significant record signal at $8\, t_S$. The horizon law therefore lives in a *quantified* domain: across the disorder range where it was measured, the ergodic sector comprises 94.5–100% of realizations, while stronger scrambling—which in this substrate also means slower transport—carries the gas into a regime

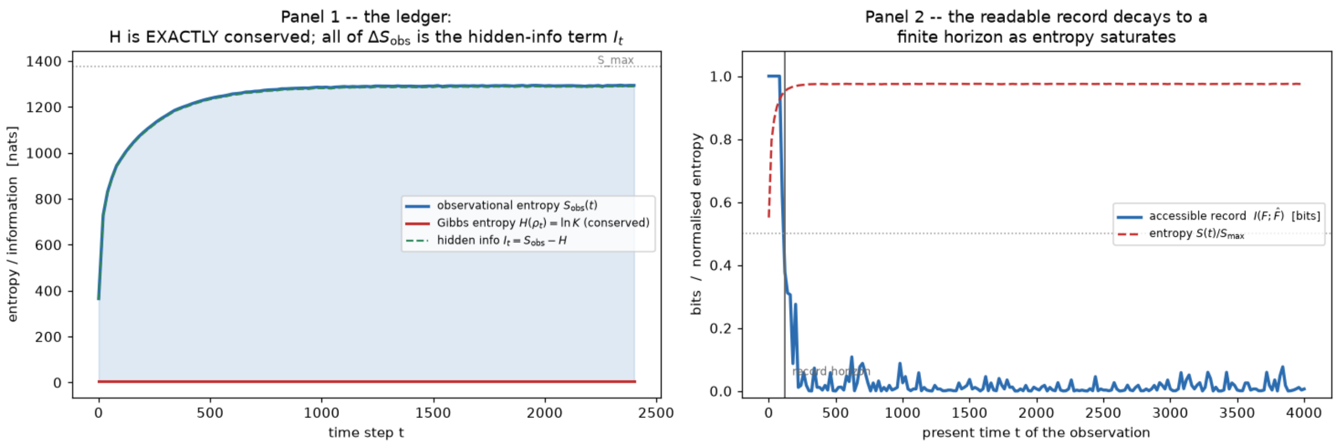

FIG. 1. **Bookkeeping** (bit-reversible lattice gas). Left ($K = 64$ worlds, $L = 128$, $b = 8$): the fine-grained entropy H of the evolving microstate ensemble is conserved to machine precision while the observational entropy $S_{\rm obs}$ rises; by the identity Eq. (4) the rise is the hidden-information term I. Right (parallel runs, $K = 64$, $L = 64$): the accessible record $I(F; \hat{F})$ of a planted one-bit fact decays from 1.000 to 0.006 bit as the gas thermalizes.

where kinetically frozen records are common. The protected sector of Sec. VII is not only engineerable; it opens spontaneously, with a measurable rate, as a function of a single disorder knob.

Re-deriving both crossings from the stored curves over a grid of gate conventions quantifies how convention-bound the *number* κ is: $\kappa = 0.87$–2.83 for $\theta_S \in \{0.85, 0.9\}$, $\theta_M \in \{0.25, 0.5, 0.75\}$ bit (a $\theta_S = 0.95$ gate lies above the finite-size equilibrium entropy and never resolves). Each individual crossing moves by $\tau\,\Delta\ln\theta$ as the crossing formula of Sec. IV prescribes—a large effect for the entropy gate because $0.9\,S_{\rm max}$ already sits close to equilibrium—which is why we treat the collapse and Eq. (9), not the numerical κ, as the convention-independent content. The law is also robust in system size: repeating the measurement with the box and marker scaled together for $L = 48$–128 ($t_{\rm S}$ spanning $6\times$) gives $\kappa(L) = 0.95$, 0.89, 0.91, 1.23, mean 0.99, with no systematic drift [27].

That κ sits slightly above unity says only that the $\frac{1}{2}$-bit record threshold is crossed slightly after the $0.9\,S_{\rm max}$ entropy threshold—a statement about conventions and amplitudes, not physics. The physical content, established next, is that both observables relax on *one* clock—and Sec. V shows exactly when that ceases to be true.

IV. A ONE-SLOW-MODE ACCOUNT

Why should the two clocks agree? In a finite ergodic system the late-time approach to equilibrium is dominated by the slowest relaxation mode; for our diffusive lattice gas this is the largest-wavelength density mode, with lifetime $\tau \sim L^2/D$. In the baseline construction both observables read that mode. The entropy deficit

$$\mathcal{D}(t) \;\equiv\; S_{\rm eq} - S(t) \;\approx\; A_S\, e^{-t/\tau_S} \tag{6}$$

measures the large-scale inhomogeneity still to be erased, while the record signal

$$M(t) \;\equiv\; I\Big(F;\hat{F}\Big)\,(t) \;\approx\; A_M\, e^{-t/\tau_M} \tag{7}$$

is the odd (left–right) component of the same relaxation: the planted fact is a parity of the slowest mode. If one timescale underlies both, $\tau_M \simeq \tau_S \equiv \tau$, the threshold crossings of Eqs. (2) and (3) give

$$t_{\rm S} = \tau \ln\frac{A_S}{\theta_S}, \qquad t^* = \tau_M \ln\frac{A_M}{\theta_M}, \tag{8}$$

with $\theta_S = 0.1\,S_{\rm max} - \mathcal{D}_{\rm eq}$ (the deficit value at the entropy gate, $\mathcal{D}_{\rm eq}$ the finite-size equilibrium deficit) and $\theta_M = \frac{1}{2}$ bit. Hence

$$\boxed{t^* = \kappa\, t_{\rm S}, \qquad \kappa = \frac{\tau_M}{\tau_S}\,\frac{\ln(A_M/\theta_M)}{\ln(A_S/\theta_S)}} \tag{9}$$

—a ratio of logarithms of measured amplitudes and chosen thresholds. Equation (9) organizes the phenomenology: (i) the proportionality itself; (ii) the flatness of κ in system size, because A_S scales with $S_{\rm max}$ so the logarithm ratio is size-free; (iii) the substrate dependence of κ (below: CA 1.08, hard disks 1.00, Clifford 0.79), because amplitudes and threshold conventions differ between substrates but enter only through slowly varying logarithms, keeping κ order one; (iv) the sensitivity to threshold choices, which enters as a $\tau\,\Delta\ln\theta$ shift of each crossing—modest when a gate sits deep in the relaxing regime, large when it approaches the equilibrium value (quantified by the gate grid of Sec. III).

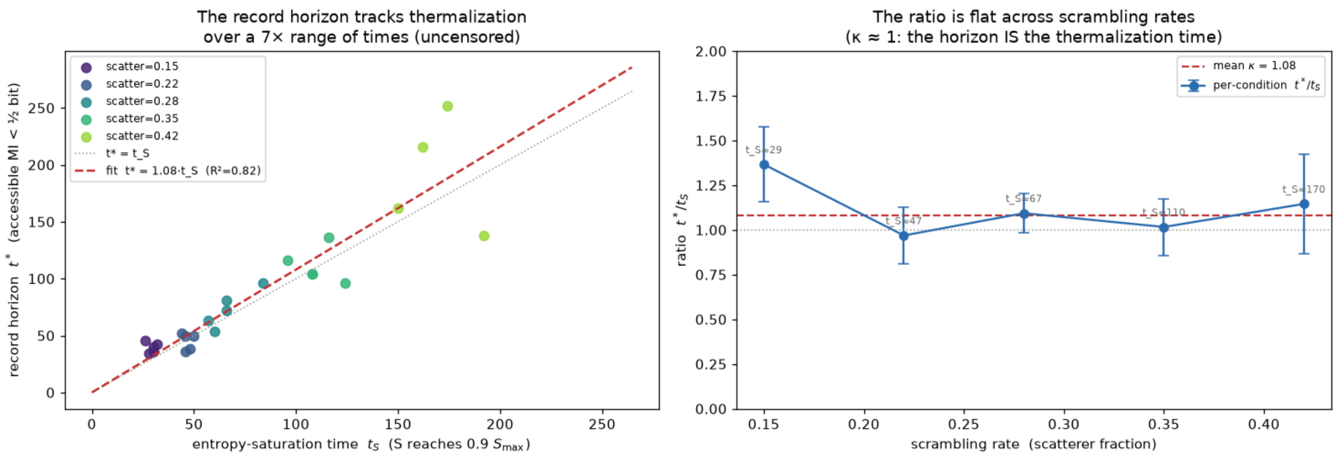

FIG. 2. **The record horizon tracks thermalization for boundary-scale records** (lattice gas; 5 scrambling rates × 5 disorder seeds; 24/25 runs resolved, nothing clamped). Left: t^* against t_S for all resolved runs; the through-origin fit gives $t^* = 1.08\,t_S$ ($R^2 = 0.82$) over a 7× range. Right: the per-condition ratio t^*/t_S is flat across scrambling rates—the law holds run by run.

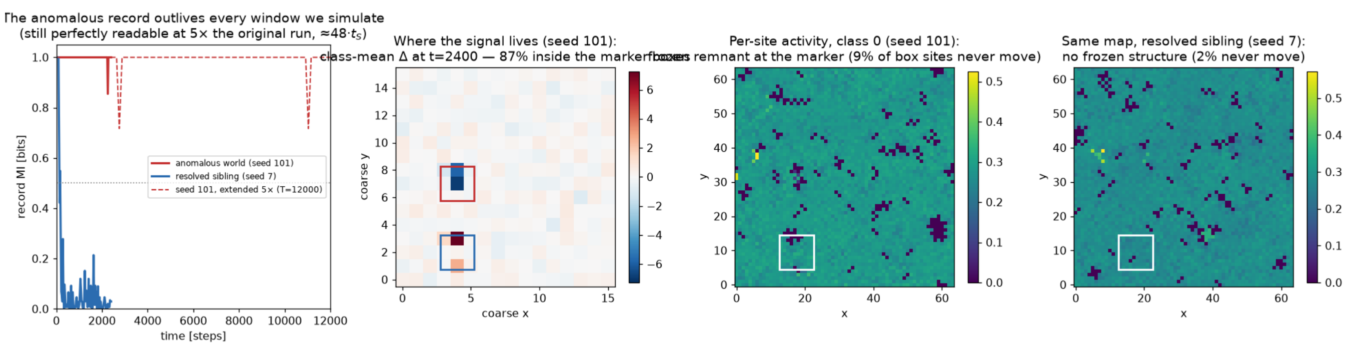

FIG. 3. **The censored run, diagnosed: a spontaneously frozen record** (scatter 0.42, one of five quenched worlds). Left: the anomalous world's record outlives every window we simulate—still perfectly readable when the run is extended to 5× the original window ($\approx 48\,t_S$)—while a resolved sibling world's record dies on schedule. Middle-left: the class-mean coarse difference at $t = 2400$ localizes at the two marker boxes (87% of $|\Delta|^2$; 81% on never-moving sites). Right two panels: per-site activity maps; the anomalous world holds a frozen solid remnant of the marker (never-move sites with occupancy 0.93 in the own box, 0.07–0.15 in the opposite, initially empty box), absent in the sibling. A fully occupied or empty 2×2 block is a fixed point of every rule within a single partition, so cluster interiors are inert and a stalled erosion stores the record for as long as we can measure ($t^* \geq 48\,t_S$; all lifetime statements are lower bounds)—behaviour consistent with the $\tau_{\text{mode}} \to \infty$ limit of Eq. (10), realized by quenched disorder.

Figure 5 tests this account directly (5 scrambling rates × 3 seeds; both tails fitted per run over their first decaying window, Appendix B). The tails are exponential with a *shared* timescale: $\tau_M/\tau_S = 0.92 \pm 0.30$ across all 15 runs, flat while the absolute τ_S varies by 7.6× (11–84 steps; condition means 0.76–1.16 with no trend). Reconstructing the crossing times from each run's fitted (τ, A) and the thresholds—no parameters beyond the fitted mode—lands at median error 5% for t^* and 1% for t_S, with the through-origin κ reconstructed at 0.98 against a measured 1.02 on the same runs. To make this a prediction rather than an in-sample reconstruction, we also predict each run's crossings from the tail parameters fitted on the *other* seeds of its condition (leave-one-out): median errors 19% for t^* and 12% for t_S. We conclude that, in this construction, record death and entropy saturation are two threshold crossings of one relaxing mode. What if the record is *not* written in the slowest mode? That is the decisive question, addressed next.

V. THE MODE-RESOLVED LAW: WHAT SETS A RECORD'S LIFETIME

The account above predicts its own generalization—and its own apparent-counterexample regime. If a

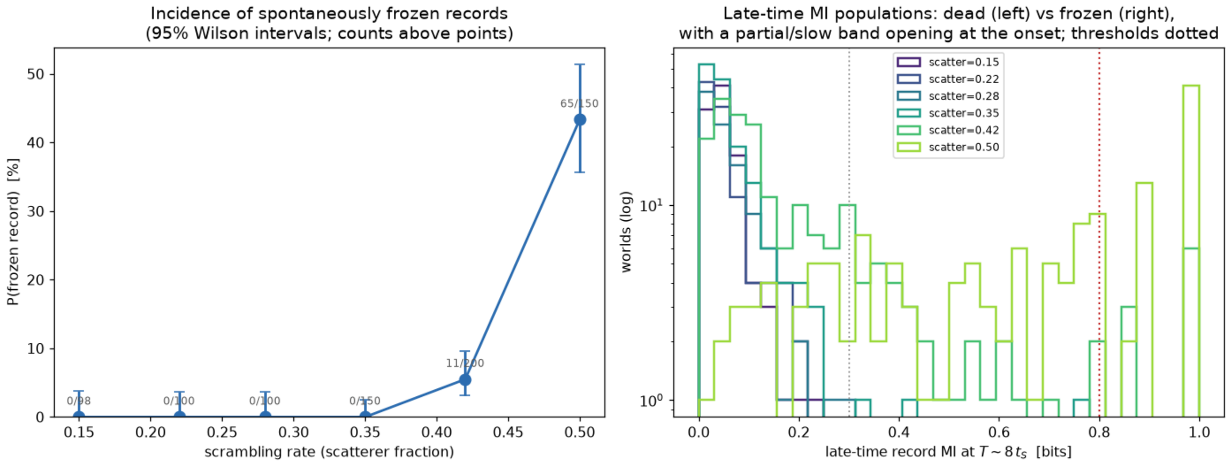

FIG. 4. **The frozen sector has a measured incidence with a sharp disorder-driven onset** (798 quenched worlds, ~ 100–200 per scrambling rate, each evolved to $T \approx 8\,t_{\mathrm{S}}$). Left: the fraction of worlds whose record is still essentially perfectly readable (late MI ≥ 0.8 bit) at $T \approx 8\,t_{\mathrm{S}}$, with 95% Wilson intervals and raw counts: 0/448 at scatter 0.15–0.35, 5.5% at 0.42, 43.3% at 0.50. Right: the late-time MI distributions behind the classification—worlds pile up dead (left peak) or frozen (right peak), with a partial/slow band opening at the onset ($0 \to 3 \to 29 \to 57$ worlds; classification thresholds dotted).

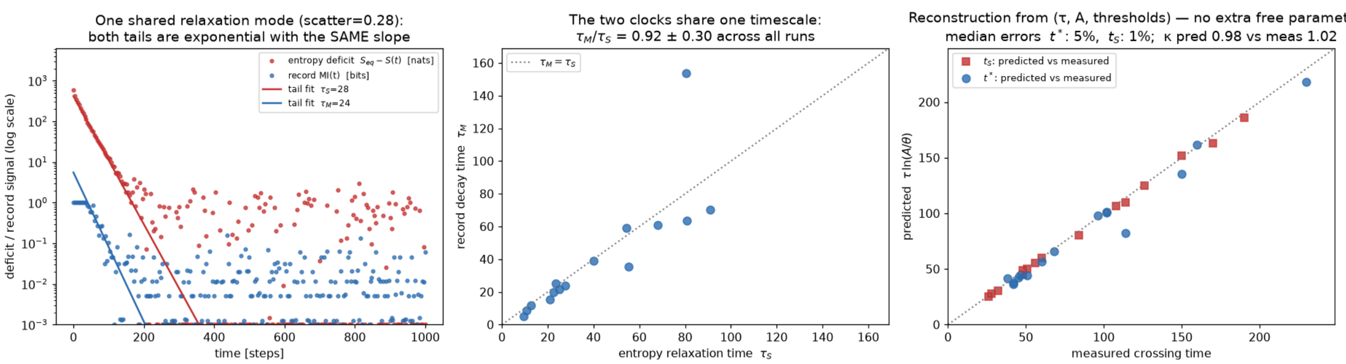

FIG. 5. **One slow mode, two thresholds** (lattice gas, 5 scrambling rates $\times$ 3 seeds). Left: for one example condition, the entropy deficit $S_{\mathrm{eq}} - S(t)$ and the record signal $I(F;\hat{F})(t)$ on a logarithmic scale; both tails are exponential with the same slope. Middle: fitted τ_M against τ_S for all runs; the ratio is 0.92 ± 0.30, flat over a $7.6\times$ range of absolute timescales. Right: reconstruction of both crossing times from the fitted (τ, A) and the thresholds, Eq. (9): median error 5% (t^*) and 1% (t_{S}); leave-one-out prediction across seeds performs comparably (see text).

record's lifetime is the threshold crossing of *the mode that carries it*, then records written at shorter wavelengths must die *before* entropy saturates, while the entropy clock, which sums over all modes, should not care which record was planted. "Record lifetime = thermalization time" would then be not a general law but the *supremum* of a mode-resolved law, attained by records written in the slowest mode. We test this by making the record's wavelength an independent knob.

The blob is prepared with an internal stripe modulation: $2m$ vertical bands, alternately at densities 0.55 and 0.35 (mean 0.45 as in all other runs), with the one-bit fact F the stripe *phase* (which band set is dense). The two classes have identical particle number by construction, and $m = 1, 2, 4, 5$ sets the record's wavelength $\lambda = w/m$. As a reference for each mode's lifetime that involves no *trained* readout we use the noise-corrected, *label-conditioned* separation power $\|\Delta(t)\|^2$ (the squared distance between the two classes' mean coarse states, with the K-world sampling floor subtracted—class labels are used, but nothing is fitted), whose decay time τ_{Δ^2} tracks the discriminating signal wherever it drifts; because the small-signal mutual information is quadratic in the mode amplitude, the MI tail time τ_M should equal τ_{Δ^2}.

Figure 6 shows the result ($K = 40$, three disorder seeds per mode):

- *The entropy clock ignores the record.* t_{S} = 98.7, 96.0, 94.7, 93.3 for $m = 1, 2, 4, 5$—flat to

2.3%, as it must be for equal-N states whose coarse entropy is dominated by the same blob relaxation.

- *The horizon is mode-matched.* $t^* = 153, 57, 21, 17$ steps—a $9\times$ spread at fixed t_S. The naive reading of Secs. III as "records live until thermalization" is therefore *false in general*: a fine-scale record is long dead while the gas is still far from saturated.

- *Each lifetime is its own mode's crossing.* Per mode, the fitted (τ_M, A_M) reconstruct $t^*(m)$ as $\tau_M \ln(A_M/\theta_M) = 154, 60, 21, 17$ — errors of 0.5%, 4.7%, 1.9%, 4.0%. The label-conditioned separation-power times agree, $\tau_M/\tau_{\Delta^2} = 0.94, 0.58, 0.74, 1.00$. (The raw t^*-vs-τ log-log slope is 1.75 rather than 1 precisely because the amplitude logarithm $\ln A_M(m)$ also varies across modes—the lifetime law has both factors.)

- *The sup form.* Only the blob-scale ($m = 1$) record survives to the entropy clock, $t^*(1) = 1.6\,t_S$; its excess over the baseline $\kappa = 1.08$ is the amplitude logarithm of the stronger stripe contrast, again as Eq. (9) prescribes.

- *Readout dependence, quantified.* At the slowest mode the three readouts agree within the logarithmic threshold offsets the mechanism predicts (fixed-template 0.82, Gaussian plug-in 1.08, relative to the trained decoder). For $m \geq 2$ the *fixed* readouts lose the record much earlier (0.19–0.41 of the trained horizon): a non-adaptive reader loses a *drifting* fine-scale pattern for physical reasons—an observation about what it takes to keep reading a record, not a decoding convention.

The general law supported by these measurements is

$$\boxed{t^* = \tau_{\text{mode}} \ln \frac{A_{\text{mode}}}{\theta_M}\,, \qquad \max_{\text{relaxing}} t^* \simeq \mathcal{O}(1)\, t_S} \tag{10}$$

where the supremum ranges over *relaxing* records in the ergodic sector and is attained by those written in the slowest thermalizing mode—which is what boundary-condition facts ("the gas started on the left") naturally are. The scope restriction is not pedantry: a non-relaxing mode escapes the bound entirely. The engineered conservation law of Sec. VII realizes the $\tau_{\text{mode}} = \infty$ limit of Eq. (10) exactly; the frozen realization of Sec. III is consistent with it on all observed timescales.

VI. CROSS-SUBSTRATE CORROBORATION

A fair objection to Secs. III–IV is that they might describe an artifact of one substrate—a square lattice gas with quenched scatterers. We therefore re-ran the two clocks, for boundary-scale records, in two deliberately different substrates. Given Sec. V, the meaningful invariant is *mode-matched* proportionality: when record and entropy relax through the same structure, t^* and t_S collapse onto one clock with order-one, convention-bound κ; when the scales are mismatched, the ratio should (and does) drift.

A. Continuous hard-disk gas

An event-driven hard-disk gas thermalizes by ballistic crossing rather than lattice scattering. The mode-matched knob is a *self-similar* size sweep: the low-entropy blob (and the planted fact) scale with the box, so record and entropy live on the same spatial scale while the one thermalization time stretches with size. Result (Fig. 7): $t^* = 1.00\,t_S$ over a $5.1\times$ range ($t_S = 14$–73), per-size $\kappa = 1.19, 1.03, 0.94, 0.97, 1.03$, and both observables collapse onto master curves under t/t_S. The complementary *mode-mismatch control* (Fig. 8) measures what happens when the scales are deliberately decoupled: the blob (and the fact) held at fixed absolute size while the box grows leaves the record on a fixed spatial scale while the entropy clock fills an ever larger volume. Measured: t_S grows $5.1\times$ across the sweep while t^* varies by only $1.4\times$, so κ drifts monotonically from 1.15 to 0.31 ($3.7\times$)—and at the smallest box, where the fixed blob coincides with the self-similar one, κ agrees with the self-similar value. The flat law is destroyed exactly when, and only when, the mode matching is broken: the mode-resolved law of Sec. V, reasserting itself in a continuous gas. (This substrate is float-chaotic, so exact values vary at the percent level with the floating-point environment; an independent re-run of the self-similar sweep on different hardware gives $\kappa = 1.04$ with the same flatness [27].)

B. Reversible quantum circuits

The bookkeeping of Sec. II is natively quantum [14, 17, 20]; the lattice gas exercises only its classical shadow. A Clifford brickwork circuit is unitary—hence exactly reversible—yet efficiently simulable [24, 25], allowing $N = 128$ qubits. Two clocks on the same random circuit: t_S is the half-chain entanglement saturation time (entanglement first reaches 0.9 of its late-time Page plateau [28, 29]); t^* is the time at which a reference qubit's recoverable record $I(R : \text{local window})$ (window $= N/3$ sites, initially 2 bits for a maximally entangled reference) first drops below 1 bit. Result (Fig. 9): $t^* = 0.79\,t_S$ with $R^2 = 0.95$ over a $6.9\times$ range ($t_S = 33$–229 layers), per-size ratios flat at 0.77–0.79 for $N \geq 64$, with the same t/t_S collapse. We note honestly that in this sweep both length scales (half chain, window) are tied to N, so a proportionality is already expected from ballistic information spreading [29]; the non-trivial content is the flat order-one value of κ across seeds and sizes, the collapse of the full curves, and the contrast with Sec. VII, where

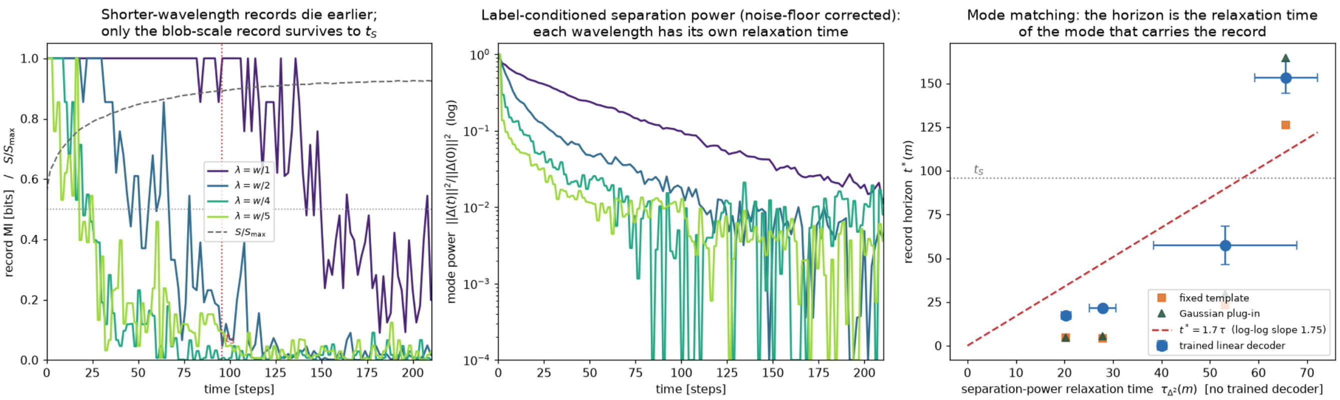

FIG. 6. **The horizon is mode-matched** (lattice gas, stripe-phase records at wavelengths $\lambda = w/m$, $m = 1, 2, 4, 5$; $K = 40$, three seeds). Left: record MI for each wavelength, with $S/S_{\max}$ (dashed) and t_S marked: shorter-wavelength records die earlier at fixed entropy clock (t_S flat to 2.3%, t^* spanning 9×). Middle: label-conditioned separation power $\|\Delta(t)\|^2$ on a log scale—each wavelength has its own relaxation time. Right: $t^*(m)$ against $\tau_{\Delta^2}(m)$ with the three readouts; each mode's lifetime is reconstructed by its own $\tau_M \ln(A_M/\theta_M)$ to 1–5% (see text).

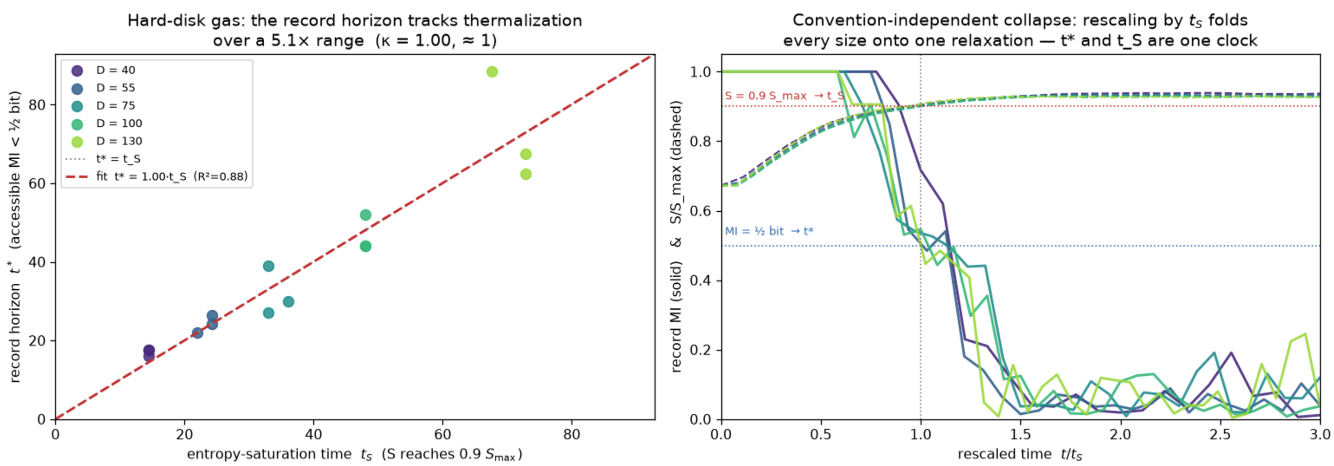

FIG. 7. **Cross-substrate corroboration I: continuous hard-disk gas** (self-similar size sweep, D = 40–130, three seed-ensembles per size). Left: t^* vs. t_S; $t^* = 1.00\, t_S$ over a 5.1× range with flat per-size ratios. Right: rescaling time by t_S collapses the record MI (solid) and the entropy relaxation (dashed) of every size onto single master curves. The complementary mode-mismatch control is measured in Fig. 8.

the identical geometry yields a *diverging* ratio once the record is protected.

TABLE I. The horizon law across substrates (boundary-scale records). κ is convention- and substrate-dependent, Eq. (9); the law is the flat proportionality.

Substrate	Dynamics	t_S range	κ
Margolus CA	bit-reversible	7×	1.08
size sweep	L = 48–128	6×	0.99
Hard-disk gas	Hamiltonian	5.1×	1.00
Clifford circuit	unitary, $N \leq 128$	6.9×	0.79
Haar statevector	unitary, $N \leq 16$	spot	0.60(10)

Clifford circuits are simulable precisely because they are non-generic, so we add a non-Clifford control: a full statevector simulation at N = 8–16 with true von Neumann entropies, run under random Clifford gates ($\kappa = 0.89 \pm 0.32$) and genuinely universal Haar-random gates ($\kappa = 0.60 \pm 0.10$). Both obey the proportional law with order-one κ; at these small sizes the absolute value is finite-size-suppressed, and the decisive point is that a universal gate set does not break the law.

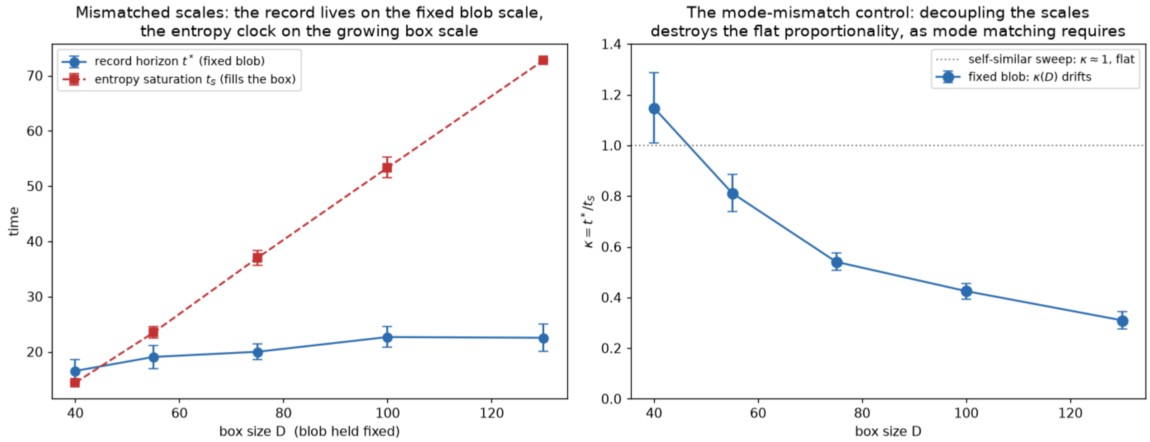

FIG. 8. **The mode-mismatch control** (hard-disk gas; blob and planted fact held at fixed absolute size while the box grows $D = 40$–130; three seed-ensembles per size). Left: the record horizon stays on the fixed blob scale ($t^* = 16.5 \to 22.5$) while the entropy clock fills the growing box ($t_S = 14 \to 73$). Right: $\kappa = t^*/t_S$ drifts monotonically $1.15 \to 0.31$ instead of staying flat; at $D = 40$ the fixed blob coincides with the self-similar one and κ matches Fig. 7. Decoupling the scales destroys the flat proportionality exactly as Eq. (10) requires.

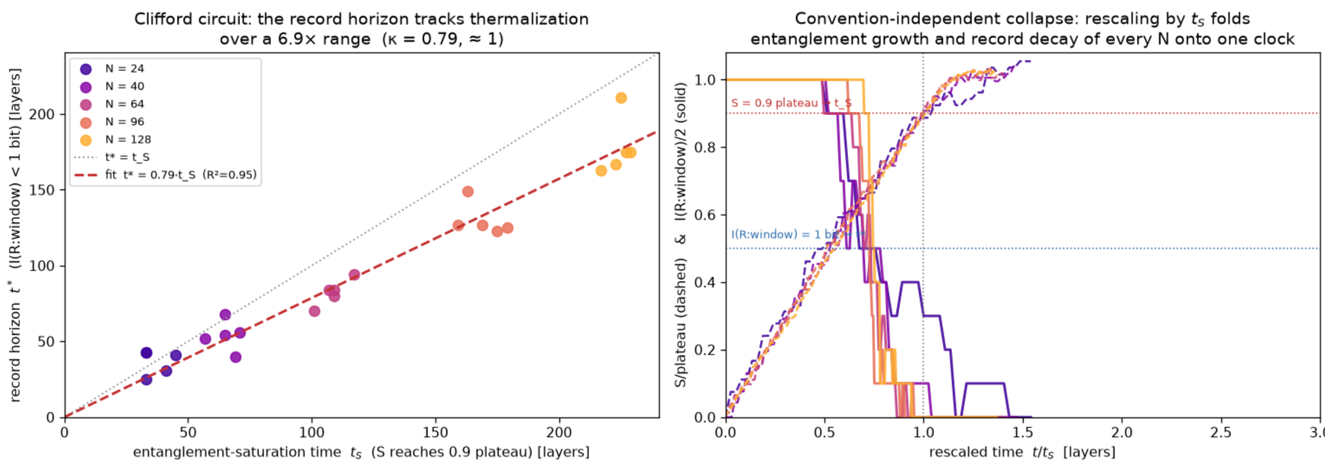

FIG. 9. **Cross-substrate corroboration II: exactly reversible quantum (Clifford) circuits** ($N = 24$–128 qubits, five circuit seeds per size). Left: record horizon vs. entanglement-saturation time; $t^* = 0.79\, t_S$ ($R^2 = 0.95$) over a 6.9× range, per-size ratios flat. Right: t/t_S collapse of entanglement growth (dashed) and record decay (solid) for all sizes. A statevector control with Haar-random gates (see text) confirms the behaviour is not a stabilizer artifact.

VII. FALSIFIABILITY: A CONSERVATION LAW DECOUPLES THE CLOCKS

A law that cannot fail is a definition. Equations (9) and (10) say the horizon tracks the thermodynamic clock only when the record is written in a *relaxing* mode; the extreme mismatch is a mode that does not relax at all. This predicts the failure regime exactly: protect the record with a conservation law, and the two clocks must decouple.

We engineer this in the Clifford substrate (Fig. 10). A reference qubit is entangled with a record qubit at the chain end. The bulk scrambles with a random brickwork (so t_S stays finite), but the bond touching the record qubit is, with probability $1 - p_{\rm break}$, a protected gate that uses the record only as a control, conserving Z_0 while still broadcasting it; with probability $p_{\rm break}$ it is a full scrambling gate. Thus $p_{\rm break}$ tunes ergodicity at the record. Result ($N = 64$, $5N$ layers, four seeds): t_S stays flat across the sweep (≈ 110 layers, relative std 0.03), while

$$\kappa(p_{\rm break}) = 0.82,\ 0.74,\ 0.80,\ 0.88,\ 1.09,\ \geq 1.67,\ \geq 2.90 \tag{11}$$

for $p_{\rm break} = 1, 0.5, 0.25, 0.1, 0.05, 0.02, 0$; at $p_{\rm break} \leq 0.02$ the record outlives every simulated window (censored—the values quoted are lower bounds), and at

$p_{\rm break} = 0$ it is never lost. The horizon law is therefore *contingent on ergodic scrambling*: a conserved quantity is exactly what lets a record outlast—or point against—the entropy gradient. This is the extreme limit of a protected (pointer/code) subspace [13]; engineered quantum memory is, in this language, the deliberate construction of horizon-law violations. Mapping this boundary is what makes the mode-matched law a falsifiable claim about generic dynamics rather than a tautology of our definitions. And the boundary is crossed spontaneously in our own data: the diagnosed censored run of Sec. III is a quenched world in which a frozen structure—no engineered gate—stores the record beyond every window we simulated ($\geq 48\, t_{\rm S}$).

VIII. REDUNDANCY: CLASSICAL DARWINISM AND ITS SCALING

How is the record stored while it lives? Following the quantum Darwinism program [12, 14, 16] we measure the partial-information profile of the classical record: the mutual information I_f recovered by decoders restricted to random fractions f of the coarse cells, and the effective redundancy $R_\delta = 1/f_\delta$, where f_δ is the smallest fraction achieving $(1-\delta) I_{\rm max}$ (we use $\delta = 0.1$; R_δ counts recoverable fragments, without asserting their mutual conditional independence). Near the boundary the profile has the classical plateau shape: 10% of the cells already recover 89% of the record, and $R_\delta \approx 8$–10, collapsing to $R_\delta = 1$ at the horizon [27]—the classical analogue of the rise and fall of redundancy [12].

A. Scaling with environment size

A diffusion/signal-to-noise argument fixes the ideal scaling. Growing the system self-similarly (box L up at fixed cell size b, marker scaled with L) adds *equivalent* fragments: each added cell is an equally informative detector of the planted parity near the boundary, so ideally one new recoverable copy per added fragment, $R \propto N$ with $N = (L/b)^2$, i.e. $\alpha = 1$ (ideal classical Darwinism). Refining the grid instead (b down at fixed L) multiplies fragments while dividing the per-fragment signal, so $\alpha < 1$ is expected—an SNR-confounded control.

Figure 11 shows the measured scaling over $N = 144$ to 16,384 ($L = 48$–512, six disorder seeds): R climbs from 7.3 to 438. A power law over the full range gives $\alpha = 0.88$; the finite-size-free tail ($L \geq 128$) gives $\alpha = 0.91 \pm 0.05$; and two independent extrapolations (a cutoff sweep in $N_{\rm min}$ and a correction fit $R = A\, N^\alpha (1 + c/N)$, mutually agreeing) give

$$\alpha_\infty = 0.92 \pm 0.04, \tag{12}$$

2.2σ below the ideal value—ideal Darwinism *supported but not certified* at accessible sizes. This quote is deliberately conservative: the data cannot distinguish the form of the finite-size correction (near-identical residuals), and the physically motivated $c/\sqrt{N}$ correction (the marker spans $\propto L = b\sqrt{N}$ cells) gives $\alpha_\infty = 0.95 \pm 0.07$, while $c\, N^{-1/4}$ gives 1.00 ± 0.12 (Appendix D). Meanwhile the SNR-confounded refinement sweep stays at $\alpha = 0.71$: the two ways of growing the environment separate cleanly, exactly as the argument predicts.

B. The quantum broadcast limit

In the stabilizer substrate we compare Zurek's dichotomy directly (Fig. 12): a pointer bit *broadcast* into N_E environment qubits (decoherence-type dynamics) yields a flat partial-information plateau—any fragment carries the full classical record—and $R_\delta = N_E$ exactly across $N_E = 8$–128, i.e. $\alpha = 1$: ideal Darwinism, realized in the noiseless limit. The same pointer under *scrambling* dynamics is encoded rather than broadcast: the partial-information profile becomes a step at half the environment and $R_\delta \approx 1$. Since perfect discrete copying has no SNR loss, the broadcast model reaches the exponent the diffusive classical lattice only approaches; this is *consistent with*—though, being a different model, it cannot by itself prove—the classical shortfall being a signal-to-noise and finite-size effect rather than a limit of the redundancy principle. (We note honestly that broadcast redundancy is fragile: a few scrambling layers collapse R_δ toward 1 well before the entropy horizon—redundant, multi-observer accessibility dies earlier than the single-copy record of Fig. 9, consistent with [12].)

IX. DISCUSSION

A. What has been established

Within exactly reversible model dynamics prepared at a low-entropy boundary: (i) the growth of coarse entropy is, by the observational-entropy identity, the growth of information hidden from the coarse observer, with the fine-grained entropy conserved to machine precision, while planted records decay to zero in parallel; (ii) for boundary-scale records the operational horizon tracks the entropy-saturation time, $t^* = \kappa\, t_{\rm S}$ with order-one κ, flat across dynamics rates and sizes and reproduced across three substrates (Table I); (iii) a one-slow-mode account reproduces the crossings in-sample and out-of-sample and reduces κ to a ratio of logarithms; (iv) resolving records by wavelength shows the general law is *mode-matched*, Eq. (10): the entropy clock is record-independent while lifetimes span an order of magnitude, each set by its own mode's relaxation and amplitude, with "lifetime = thermalization time" as the supremum attained by slowest-mode records; (v) the law fails precisely when the record is stored in a non-relaxing structure—the $\tau \to \infty$ limit of the same statement—whether engineered (a conservation law, Sec. VII) or spontaneous, and the sponta-

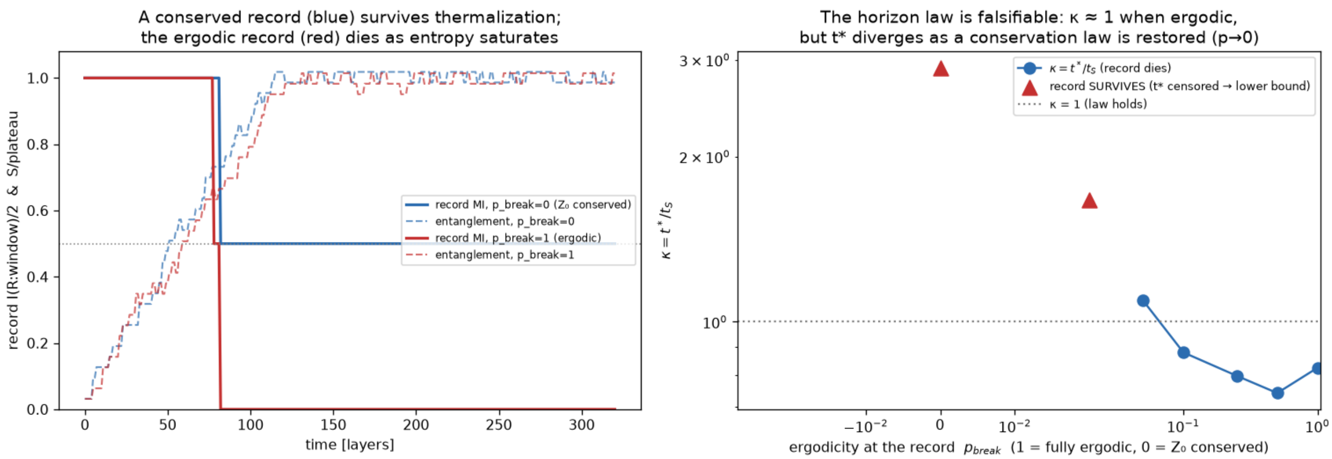

FIG. 10. **The law fails where it must** (Clifford chain, $N = 64$; a tunable Z_0-conserving gate protects the record with probability $1 - p_{\mathrm{break}}$). Left: example trajectories—under full scrambling the record dies as entanglement saturates, while the protected record survives arbitrarily long although the bulk thermalizes identically. Right: $\kappa = t^*/t_S$ against p_{break}; the ratio is order one in the ergodic regime and diverges (censored points are lower bounds) as the conservation law is restored, while t_S stays flat.

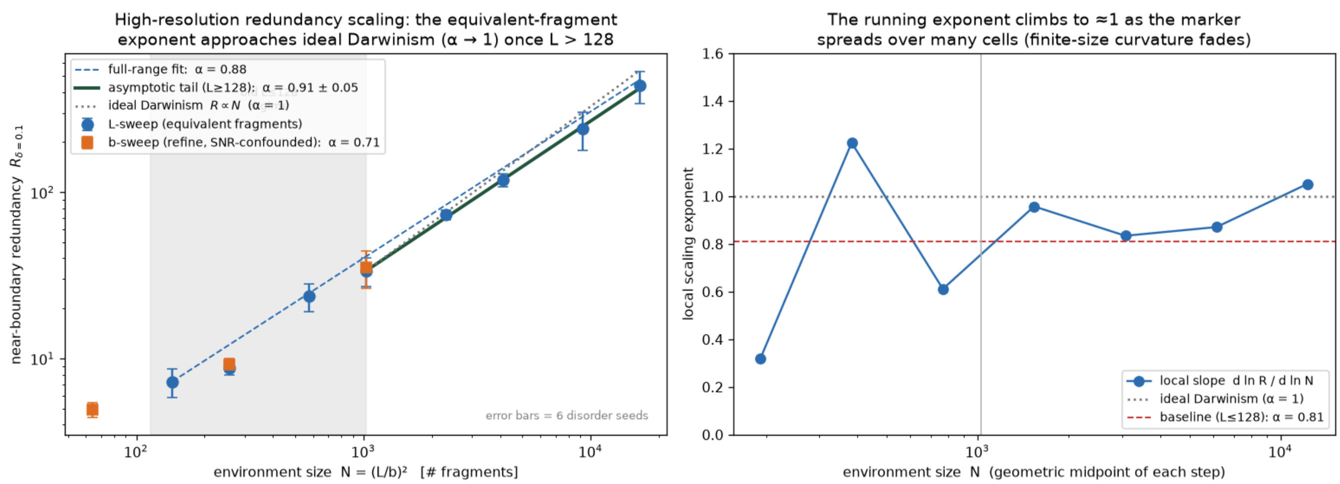

FIG. 11. **Redundancy scaling toward ideal Darwinism** (lattice gas, equivalent-fragment sweep $L = 48$–512, i.e. $N = (L/b)^2$ up to 16,384; six disorder seeds). Left: near-boundary effective redundancy $R_{\delta=0.1}$ vs. N; the full-range exponent is 0.88, the $L \geq 128$ tail 0.91 ± 0.05 (extrapolating to $\alpha_\infty = 0.92 \pm 0.04$, conservatively; see text), while the SNR-confounded refinement sweep stays at $\alpha = 0.71$. Right: the local slope $d \ln R / d \ln N$ climbs to ≈ 1 at the largest sizes as the small-L curvature fades.

neous sector has a measured, sharply-onsetting incidence in disorder strength ($0/448 \to 5.5\% \to 43\%$, Sec. III); (vi) records are stored redundantly, with an effective-redundancy scaling that approaches ideal classical Darwinism and attains it in the noiseless quantum broadcast.

Taken together, these results sharpen the qualitative account of the memory arrow [6–9]. Mlodinow and Brun argued that a robust record-forming system must sit on the entropy-increasing side of the gradient [8]; Rovelli, that memories degrade as the system thermalizes [9]. Our measurements supply the quantitative structure: *which* records last is set by mode matching; the most durable record's lifetime is the thermalization time itself; objectivity while alive is redundancy; and outliving the gradient requires a non-relaxing structure. In our data that happened two ways: by engineered protection (Sec. VII), and unbidden, in one of 25 disordered realizations, by spontaneous freezing (Sec. III)—nature does occasionally build its own memory protection, and when it does, the record decouples from the thermodynamic clock for as long as we can follow it. In companion experiments in the same testbed [27], the *direction* of record accumula-

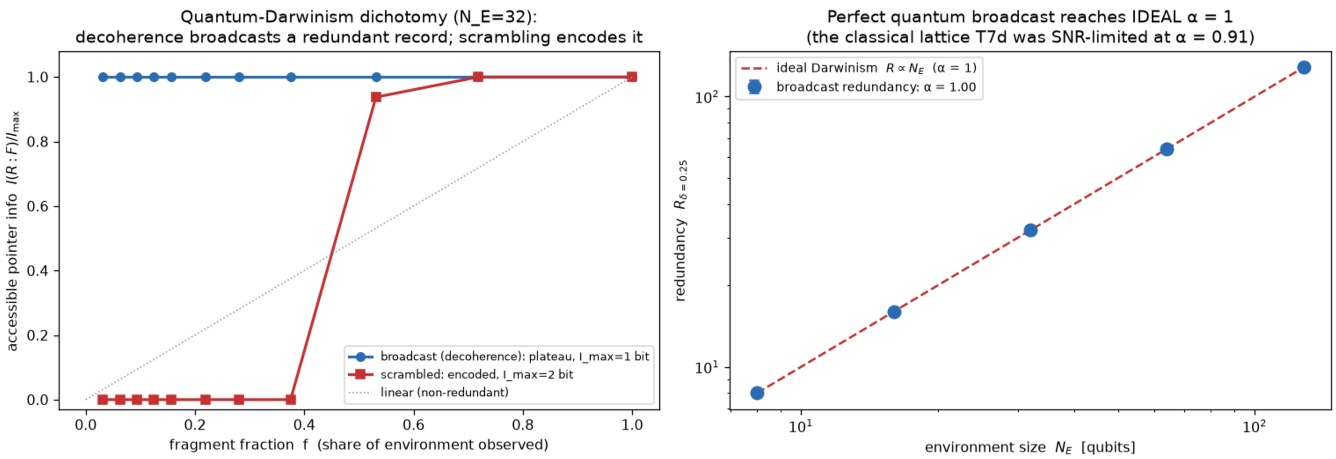

FIG. 12. **Quantum Darwinism in the reversible circuit substrate.** Left: partial-information profiles for a pointer bit broadcast into the environment (flat plateau: any small fragment carries the record) versus scrambled into it (step at half the environment: the record is encoded, not redundant). Right: effective redundancy of the broadcast record scales as $R_\delta = N_E$ exactly ($\alpha = 1$, $N_E = 8$–128)—the ideal limit the classical lattice approaches from below.

tion flips with the sign of the entropy gradient (including for facts injected mid-run, realizing Reichenbach's fork asymmetry), completing the picture: direction, lifetime, and storage of memory are all read off the gradient.

B. What has *not* been shown

First and fundamentally, nothing here derives the arrow of time: every run assumes the low-entropy boundary. This is not a defect of the testbed but a theorem-level wall for any time-symmetric dynamics [4, 26]; our exactly reversible substrate makes it vivid, since entropy grows in *both* time directions away from the boundary state [27]. The mystery of the Past Hypothesis is used, not dissolved. (An exploratory companion experiment relocates the posit in the spirit of Carroll and Chen [30]—a *small* early volume plus expansion, rather than a fine-tuned microstate, sustains an unbounded two-headed entropy valley from a single bounce microstate evolved in both time directions—but the relocated question, why the volume was small, remains as open as the original [27].)

Second, t^* is operational: the horizon of an explicit decoder class, hence a lower bound on the accessible-information horizon. We quantified its readout dependence (Sec. V) but have not computed the information-theoretic horizon itself. Third, κ's numerical value is convention-laden; only the shared relaxation, its mode structure, and the collapses are physics. Fourth, the classical redundancy exponent is supported at, but not certified to, the ideal value ($\alpha_\infty = 0.92 \pm 0.04$ conservatively, up to 1.00 under equally consistent extrapolation forms); certification would require another decade in N. Fifth, all claims are about the model classes simulated here—though three substrates, a mode-resolved mechanism, and a mapped failure boundary argue that mode-matched horizons are a generic property of ergodic reversible information dynamics rather than an artifact of any one model.

C. Outlook

Three extensions seem most valuable. (i) A quantitative tie of the mode times to transport: the account predicts $\tau(\lambda) \propto \lambda^2/D$ in the hydrodynamic regime, with D set by the scatterer density; our measured $\tau_{\Delta^2}(m)$ decays more slowly with m than λ^2 at these sizes, and mapping that crossover would express the horizon entirely in transport coefficients. (ii) Computing the information-theoretic horizon (or tighter decoder hierarchies) to close the operational gap. (iii) In the corroboration direction [27]: conditioning equilibrium fluctuations on reproducing one half of a record and testing whether the other half corroborates would quantify why multiple independent records justify belief in a real low-entropy past—Reichenbach's common cause as a measurable statement. (iv) The measured onset of the frozen sector invites a mechanism study: what structural change controls it, how the onset scales with marker geometry and system size, and how this classical sector relates to the broader physics of weak ergodicity breaking and fragmentation-protected memory.

ACKNOWLEDGMENTS

All simulations, per-experiment numerical verdicts, the scripts generating every figure, and per-run data for

the core experiments (`data/`) are openly available [27].

Appendix A: The bit-reversible substrate

The Margolus automaton partitions the $L \times L$ periodic grid into 2×2 blocks, with the partition offset alternating each step. Each block is updated by a bijective lookup table acting on its 16 possible occupancies; the composition of bijections is a bijection, so the global map is exactly invertible, and running the inverse tables in reverse order returns the initial bitstring exactly (verified bit-for-bit [27]). A quenched fraction (the scrambling knob) of blocks applies a rotation rule; the remainder apply the identity-preserving collision rule. Initial low-entropy states confine all particles to a width-w blob at density ρ ($w = 40$, $\rho = 0.45$, $L = 64$, $b = 4$ for the horizon experiments), with microstates sampled microcanonically. The hard-disk companion is an event-driven elastic-disk gas in a square box with specular walls; between collisions the flight is exact, so the only irreversibility is floating-point.

Appendix B: Records, decoders, and tail fits

The baseline fact F places a marker of identical particle number on the left or right of the blob, so the classes share every conserved quantity; the mode-resolved facts of Sec. V are stripe phases at wavelength w/m, also with identical N by construction. For each class, K microstates are evolved through the same quenched disorder ($K = 40$ for the horizon and mode sweeps, 64 for the ledger). At each sample time the coarse occupancy vectors of half the worlds train a nearest-centroid linear readout $\hat{F}$; $I(F;\hat{F})$ is computed from the held-out confusion matrix. This makes t^* an *operational* horizon: the time past which this readout of the present coarse state retains less than half a bit about the fact. The alternative readouts of Sec. V are (a) an untrained fixed-template decoder (the sign of the projection onto the $t = 0$ stripe pattern) and (b) a Gaussian plug-in mutual information on the held-out one-dimensional stripe projection, with the finite-sample separation bias $\sigma^2(2/n)$ subtracted so the estimate decays to zero rather than to a sampling floor. The label-conditioned separation power is $\|\Delta(t)\|^2 = \|\bar{V}_0(t) - \bar{V}_1(t)\|^2$ minus its K-world sampling floor, where $\bar{V}_c$ is the class-mean coarse state (class labels enter; no decoder is trained).

For the ledger, an ensemble of K equiprobable microstates gives $H = \ln K$ exactly under bijective evolution, while the coarse observer assigns the observational entropy [17, 19]; the hidden-information term of Eq. (4) is the difference, computed exactly from the evolving ensemble.

For the tail fits, the equilibrium deficit $\mathcal{D}_{\rm eq}$ (mean of the last 15% of each run) is subtracted before fitting; each tail is fitted by ordinary least squares on $\ln y$ over its first decaying crossing of a fixed window ($[0.04, 0.5]$ of the initial deficit for $\mathcal{D}$; $[0.04, 0.6]$ bit for M), which excludes the estimation floor that follows the record's death.

Appendix C: Stabilizer methods

Clifford states are tracked by the binary check matrix of their stabilizer group (phases dropped; they do not affect entanglement). Gates act as column operations [24, 25]. The entanglement entropy of a region A is $S_A = \mathrm{rank}_{\mathbb{F}_2}(G|_A) - |A|$ ebits [31], and mutual informations follow from $I(A{:}B) = S_A + S_B - S_{AB}$. Brickwork layers apply independent random two-qubit Cliffords (random single-qubit words interleaved with CNOTs in both directions) to alternating bonds; this drives contiguous regions to volume-law (Page) entanglement [28, 29]. The protected gate of Sec. VII uses the record qubit only as a control, conserving Z_0. The statevector control applies exact 4×4 unitaries (Haar-random via QR of complex Gaussians, or random Clifford words) and computes von Neumann entropies from Schmidt spectra.

Appendix D: Extrapolation of the redundancy exponent

Two estimators of the asymptotic exponent agree: (A) a cutoff sweep, fitting $\alpha_{\rm eff}(N_{\rm min})$ on $N \geq N_{\rm min}$ and extrapolating linearly in $1/N_{\rm min} \to 0$ (giving 0.918); and (B) a correction fit $\ln R = c_0 + \alpha_\infty \ln N + c_1 N^{-p}$ with seed-bootstrapped errors. For $p = 1$: $\alpha_\infty = 0.922 \pm 0.036$; $p = \frac{1}{2}$: 0.947 ± 0.067; $p = \frac{1}{4}$: 1.001 ± 0.124—with statistically indistinguishable residuals (RSS 0.060/0.063/0.064), so the main text quotes the most conservative value. The local slope $d \ln R/d \ln N$ reaches 0.96–1.05 across the largest size steps.

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